18th Iranian Statistics Conference
K.N.Toosie University of Thechnology
29-31 August 2026

Order Statistics in R: Moment Computation, Goodness-of-Fit Assessment, and Statistical Inference with the mos Package

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Abstract:

Although order statistics are fundamental in statistical inference, reliability analysis, and goodness-of-fit assessment, computational tools for their moment evaluation remain limited. In this paper, we briefly introduce the R package mos which provides a unified framework for the analysis of order statistics, censored samples, and k-record values. The package combines analytical evaluation of moments, whenever closed-form expressions are available, with Monte Carlo estimation for more general distributions. It also provides functions for variance, skewness, kurtosis, and efficient simulation of ordered-data structures. Applications to graphical goodness-of-fit diagnostics and best linear unbiased estimation illustrate the practical utility of the package. The mos offers a consistent and extensible environment for research and applications involving ordered data.

Keywords: Censoring, Order statistics, Moments, Record values, Reliability, R programming language.

Mathematics Subject Classification (2010): 62G30, 62-04, 62N05.

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1 Introduction

Order statistics, obtained by arranging a random sample $X_1, \ldots, X_n$ in non-decreasing order as $X_{1:n} \leq \cdots \leq X_{n:n}$, are fundamental in statistical theory and applications. They support extreme-value analysis through sample minima and maxima, provide robust estimators such as medians and trimmed means, and naturally arise in reliability and survival data settings (David and Nagaraja, 2003). Closely related data structures include censored samples, in which only partial order information is available under schemes such as Type-I, Type-II, or progressive censoring, and record values, which arise as successive extremes in sequential observations and naturally extend to k-record processes. A central computational challenge in working with order statistics is the evaluation of their moments. Single and product moments are essential in methods such as best linear unbiased estimation (BLUE), L-moments and goodness-of-fit procedures (Arnold et al., 2008). While closed-form expressions exist for several parent distributions, they are scattered across the literature and lack a unified computational implementation. For many commonly used distributions, such expressions are unavailable, and simulation-based approximations are required.

Several R packages address individual components of this workflow. Existing tools provide functionality for generating order statistics, handling censored data, or computing L-moments. However, most focus on isolated tasks. To the best of our knowledge, no existing package provides a unified framework that integrates analytical and simulation-based computation of general order-statistic moments with tools for censored samples and record-value generation within a single coherent interface. The mos package (Arafeh and Salehi, 2025), provides a unified computational framework for order-statistic analysis. It implements both analytical and simulation-based computation of moments, depending on the availability of closed-form expressions in the literature. The package further offers tools for deriving higher-order summaries such as variance, skewness, and kurtosis, and includes efficient simulation routines for order statistics, censored samples, and k-record values within a consistent interface.

The remainder of the paper is organized as follows. Section 2 describes the computational framework of package. Section 3 illustrates usage examples, and Section 4 presents applications to goodness-of-fit assessment and statistical inference. Section 5 concludes the paper.

2 Computational framework of the mos package

Let $X_1, \ldots, X_n$ be a continuous random sample drawn from a cumulative distribution function (cdf) F with density f, and let $X_{r:n}$ denote the rth order statistic. Its kth single moment (David and Nagaraja, 2003) is

$$\mu_{r:n}^{(k)} = \mathbb{E}\big[X_{r:n}^k\big] = r\binom{n}{r}\int_{-\infty}^{\infty} x^k\,[F(x)]^{r-1}[1-F(x)]^{n-r}f(x)\,\mathrm{d}x$$

The mos exposes $\mu_{r:n}^{(k)}$ through the mo_*() functions using a dual framework: whenever a tractable closed form is available it is evaluated directly (deterministic and fast), and otherwise the moment is estimated by Monte Carlo. Closed forms are used for the uniform, exponential, Weibull, symmetric triangular, Topp–Leone and complementary beta distributions (with references in Table 1); Monte Carlo is used for the normal, Student t, gamma, beta, Pareto and Kumaraswamy (Kumaraswamy, 1980) parents, with defaults $M = 10^5$ replicates and seed $= 42$. The variance, skewness and the Pearson measures of kurtosis of $X_{r:n}$ follow from the first four single moments and are returned by varOS(), skewOS() and kurtOS(). Table 1 summarizes the exported functions.

Table 1: Main exported functions of mos. The analytical mo_*() functions evaluate closed-form moments from the cited sources; the remaining mo_*() functions use Monte Carlo.

Function	Purpose
ros()	Simulates order statistics (single or several ranks)
rcens()	Simulates Type-I and Type-II censored samples
rpcens2()	Simulates progressive Type-II censored samples
rkrec()	Simulates first upper / lower k-record values
mo_unif()	Uniform moments (Arnold and Balakrishnan, 2012)
mo_exp()	Exponential moments (Ahsanullah et al., 2013)
mo_weibull()	Weibull moments (Harter and Balakrishnan, 1996)
mo_tri()	Symmetric triangular moments (Nagaraja, 2013)
mo_topple()	Topp–Leone moments (Genç, 2012)
mo_compbeta()	Complementary beta moments (Jones, 2002; Makouei et al., 2021)
mo_norm(), mo_t(), mo_gamma(), mo_beta(), mo_pareto(), mo_kumar()	Normal, t, gamma, beta, Pareto, Kumaraswamy (Monte Carlo)
varOS(), skewOS(), kurtOS()	Variance, skewness, kurtosis of $X_{r:n}$

All simulation rests on the generator ros(). Since $F(X_{r:n}) \sim \mathrm{Beta}(r, n-r+1)$ for any continuous

F, a single rank is drawn as $F^{-1}(B)$ with $B \sim \text{Beta}(r, n-r+1)$—one beta draw and one quantile evaluation, with no sorting—so order statistics can be generated from any continuous distribution with a known quantile function, built in or supplied through qf; for several ranks ros() generates, sorts and subsets complete samples to preserve their joint distribution. mos also supports two related ordered-data structures. Censored samples (Balakrishnan and Aggarwala, 2000) arise in life-testing: Type-I stops the test at a fixed time T, Type-II at the rth failure (the first r order statistics), and progressive Type-II withdraws R_i surviving units at the ith failure for a scheme $R = (R_1, \ldots, R_m)$ with $m + \sum_i R_i = n$; these are simulated by rcens() and rpcens2() (the latter using the algorithm of (Balakrishnan and Sandhu, 1995)). k-record statistics (Arnold et al., 1998) generalize ordinary records by tracking the kth largest value seen so far (the case $k = 1$ is the usual record); the first upper and lower k-record values are generated by rkrec().

3 Using the package

The generator ros() simulates order statistics: a scalar r returns one rank and a vector r a matrix preserving their joint distribution. It also accepts a user-defined quantile function through the qf argument, so order statistics can be drawn from any continuous distribution. The mo_*() functions return single moments through a common interface (deterministic for the analytical families, Monte Carlo otherwise), and varOS(), skewOS(), kurtOS() return the corresponding summaries:

```
> set.seed(1)
> ros(5, r = 2, n = 10, dist = "norm")
[1] -1.238927 -0.808177 -1.348781 -0.726418 -1.340785
> qgompertz <- function(p, a, b) log(1 - log(1 - p) * b / a) / b
> set.seed(2)
> ros(5, r = 3, n = 8, qf = "qgompertz", a = 1, b = 2)  # user quantile fn
[1] 0.1760290 0.3252047 0.1487873 0.2850333 0.3170499
> mo_unif(r = 2, n = 5, k = 1)   # closed form = r/(n+1) = 1/3
[1] 0.333333
> mo_norm(r = 2, n = 10, k = 4)  # Monte Carlo
[1] 2.549317
```

Censored samples and k-record values are generated analogously through rcens(), rpcens2() and rkrec(); for example, a progressive Type-II sample with scheme $R = (2, 1, 2, 0, 0)$:

```
> set.seed(1)
> rpcens2(n = 10, R = c(2, 1, 2, 0, 0), dist = "exp", rate = 1)
[1] 0.1601063 0.1738609 0.2852860 0.7795502 2.1056580
```

4 Application

We illustrate two applications of order-statistic moments computed with mos: a graphical goodness-of-fit diagnostic for the complementary beta distribution (Section 4.1), and best linear unbiased estimation of location and scale parameters (Section 4.2).

4.1 Goodness-of-fit for the complementary beta distribution

Expected order-statistic values furnish a graphical goodness-of-fit diagnostic for distributions, as exploited by Makouei et al. (2021). More specifically, for an ordered sample $x_{(1)} \leq \cdots \leq x_{(n)}$ one plots the estimated values of $\mathbb{E}[X_{i:n}]$ against the sorted observations $x_{(i)}$. Under a well-fitting model, the points lie near the identity line, whereas systematic departures indicate lack of fit. We reproduce the flood-levels application of Makouei et al. (2021) using mos.

The data are the maximum flood levels (in millions of cubic feet per second) of the Susquehanna River at Harrisburg, Pennsylvania, over twenty four-year periods from 1890 to 1969:

```
> flood <- c(0.654, 0.613, 0.315, 0.449, 0.297, 0.402, 0.379, 0.423,
+            0.379, 0.324, 0.269, 0.740, 0.418, 0.412, 0.494, 0.416,
+            0.338, 0.392, 0.484, 0.265)
```

Following Makouei et al. (2021), the complementary beta (CB) distribution is compared with the beta and Kumaraswamy (Kw) distributions, using their L-moment parameter estimates (Table 2). For each model the expected means $\mathbb{E}[X_{i:n}]$, $i = 1, \ldots, 20$, are computed with mo_compbeta() (closed form, CB), mo_beta() and mo_kumar() (Monte Carlo).

Figure 1 plots the expected means against the sorted data for all three models. Plotting them against the sorted data, the CB and beta points cluster on the identity line whereas the Kw points depart from it in the lower tail, agreeing with the KS and AD tests given by Makouei et al. (2021).

4.2 Best linear unbiased estimation of location and scale

A classical use of order-statistic moments is the construction of best linear unbiased estimators (BLUEs) of the location and scale parameters of a location–scale family based on its order

Table 2: The L-moment parameter estimates and goodness of fit criteria associated with the Susquehanna River flood-levels data computed using the mos packages, as reported by (Makouei et al., 2021, Table 8).

Model	$\hat{a}$	$\hat{b}$	KS p-value	KS test statistic	AD p-value	AD test statistic
CB	0.22	0.16	0.475	0.189	0.593	0.685
Beta	6.69	9.12	0.466	0.189	0.559	0.696
Kw	1.67	3.21	0.045	0.307	0.067	2.247

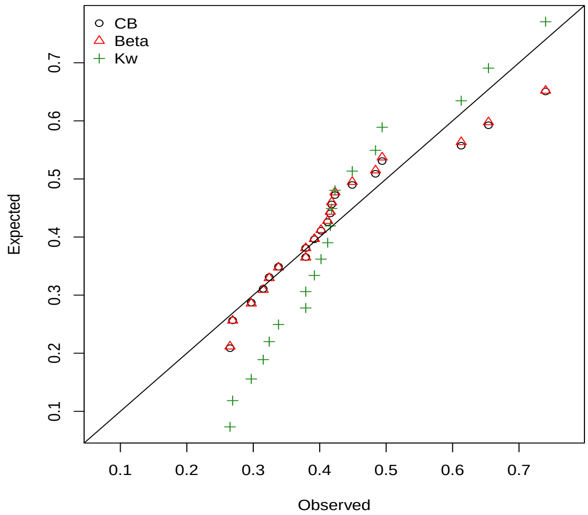

Figure 1: Expected order-statistic means $\mathbb{E}[X_{i:20}]$, $i = 1(1)20$, versus the sorted Susquehanna River flood-levels data.

statistics, i.e. $X_{i:n} = \mu + \sigma Z_{i:n}$, where Z is the standardized parent. Writing $\boldsymbol{\alpha} = \mathbb{E}[\mathbf{Z}]$ for the vector of standardized order-statistic means, $\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{Z})$ for their covariance matrix, and $\mathbf{X} = (X_{1:n}, \ldots, X_{n:n})^\top$, the BLUEs of the location and scale parameters are respectively obtained as (Lloyd, 1952; Balakrishnan and Li, 2008)

$$\hat{\mu} = -\boldsymbol{\alpha}^\top \boldsymbol{\Gamma} \mathbf{X}, \qquad \hat{\sigma} = \mathbf{1}^\top \boldsymbol{\Gamma} \mathbf{X}, \qquad \boldsymbol{\Gamma} = \frac{\boldsymbol{\Sigma}^{-1}(\mathbf{1}\boldsymbol{\alpha}^\top - \boldsymbol{\alpha}\mathbf{1}^\top)\boldsymbol{\Sigma}^{-1}}{(\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1})(\boldsymbol{\alpha}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\alpha}) - (\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\alpha})^2}, \tag{4.1}$$

where $\mathbf{1}$ is the vector of ones. The inputs $\boldsymbol{\alpha}$ and $\boldsymbol{\Sigma}$ are exactly the order-statistic moments mos delivers: the means $\boldsymbol{\alpha}$ from the mo_*() functions and the variances $\mathrm{Var}(Z_{r:n})$ on the diagonal of $\boldsymbol{\Sigma}$ from varOS(); since product moments are not yet implemented, the off-diagonal covariances are obtained by simulating the full ordered sample with ros().

We illustrate this using the remission times (in years) of 20 leukemia patients, modelled by a two-parameter exponential distribution as in Barakat et al. (2021), where a Kolmogorov–Smirnov test indicates a good fit ($D = 0.103$, $p = 0.97$). The quantities required for linear estimation are obtained directly from mos: expected order statistics via mo_exp(), variances via varOS(), and covariances through simulation with ros(). These moments can then be assembled to construct the BLUEs of the location and scale parameters. For the two-parameter exponential distribution, the BLUEs are also available in closed form (Arnold et al., 2008)

$$\hat{\mu} = \frac{nX_{1:n} - \bar{X}}{n-1}, \qquad \hat{\sigma} = \frac{n(\bar{X} - X_{1:n})}{n-1},$$

providing a convenient benchmark against which the moment-based estimates can be compared. Applying both approaches to the leukemia data yields identical estimates, $\hat{\mu} = 0.95$ and $\hat{\sigma} = 1.24$, with any negligible discrepancies attributable to Monte Carlo error in the simulated covariance matrix. In contrast, the ordinary sample mean $\bar{X} = 2.19$ estimates $\mu + \sigma$ rather than the location parameter μ and is not consistent for μ, illustrating the advantage of BLUE-based inference for this asymmetric family. This example shows how the moment-computation core of mos—means from mo_exp(), variances from varOS(), and covariances via ros()—feeds directly into classical linear estimation; built-in BLUE support and product moments are a natural direction for further development.

5 Discussion

The mos package unifies the moments computation and simulation of order statistics, censored samples, and k-record values within a single framework. By combining exact formulas with Monte Carlo methods, it supports both theoretical investigations and practical data analysis. Future work will focus on product moments, direct covariance computation, and dedicated routines for BLUE and related inferential procedures.

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